

Funky Conditions

Canada IMO Camp

July 2023

This lecture is about decoding geometry conditions which are difficult to construct. This is *not* a lecture about how to solve geometry problems with difficult conditions; rather it is about the conditions themselves.

This is mostly a lecture about methods, rather than concrete theory. As a result I will be somewhat unrigorous. Please feel free to ask me any questions that arise (from this lack of clarity, or from other reasons).

Problem citations are not given in text because I personally find this has a detrimental effect on the reader's mindset in solving the problem. But they are given in comments in the latex file.

1 Constructions

Oftentimes many geometry conditions can be decoded by “moving” angles and lines until you can relate them with other angles / lines. There are plenty of ways to do this, and this is an art which has no fixed, easily describable technique. Nevertheless here are some ideas to get you started:

1. If you're given an angle $\angle ABC$ which you want to relate to some other angle $\angle ADE$, consider intersecting (ABC) with AD , so you can now compare $\angle ADC$ and $\angle ADE$. In general it will be very useful to use the circumcircle of a triangle to move its angles around.
2. Consider also constructing parallel lines and similar triangles to move your angles and lengths around.
 - (a) Along this vein, it's often also useful to try to construct similar triangles which share an orientation, and thus consider the *spiral similarity* connected them (such a spiral similarity exists and is always unique; why?) Spiral similarities have a wealth of useful properties, and so finding good spiral similarities to connect your points is often useful.
3. Reflections are also always useful.
4. If you want to compare two side lengths, consider the circles which have one of the side lengths as a radius.

Here is a starting example.

Example 1.1. Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

As a starting hint: let's ignore the final concurrency for now and just work with the condition. What constructions would help us compare the angles $\angle BC_1C$, $\angle CA_1A$, and $\angle AB_1B$?

Here is an example which basically forces you to think about constructions.

Example 1.2. Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

Here's a walkthrough.

1. The strange length condition is basically staring us in the face. What constructions might help us get rid of it?
2. Let's call $\omega = (ABCD)$, and denote $\omega_A = (A, AD)$, $\omega_B = (B, BC)$. This kind of helps, but we are still faced with the strange condition $\angle APD = \angle BPC$ for the point P . How can we construct the point P ?
3. Let's denote reflections of points across BC using primes $'$. Notably, DC' intersects ω_B at a second point K . There are many strange things you can notice from this concurrence, but here are two obvious ones (provided you have been drawing with straightedge and compass):
 - (a) K also lies on AC ;
 - (b) K' lies on $(ABCD)$.

Let's start with the first item. Prove that this concurrence is equivalent to $AK \cdot AC = AD^2$.

4. Now, prove that this identity is equivalent to $(ABCD)$ and DP are inverses with respect to ω_A . Notably, P should be the inverse of B with respect to ω_A !
5. We can try and prove now that this inversive relation is actually true, but we can also phantom point. Define now P to be this inverse, and prove that it satisfies all the given properties (and is also the unique point to do so).
6. There are many ways to finish from here. For example, compute AP/PB now in terms of the sides of $ABCD$, and use sines to finish.

One note about constructions: the constructions we perform to solve the problem are never frivolous. We don't construct similar triangles, parallel lines, or circles "just because they might help"; we do them because we have an explicit relation between two geometrical quantities that we want to decode, and we use the most intuitive constructions to decode them. In short, always keep your eye on the ball.

Remark 1.3. A brief sidenote on constructing good diagrams: there is a lot of debate whether it is important to construct perfect diagrams, or if it's not really all that necessary. Everyone does geometry differently so there is not one fixed answer. But in my personal experience, constructing good diagrams is useful not always because it lets you see coincidences very precisely, but also because the *process* of constructing a diagram is often very useful. To make a perfect diagram, you need to choose well which points to start with, and know how to construct every succeeding point with straightedge and compass. This can be a difficult task; even more so if you are trying to make the construction quick. This forcing yourself to think about constructions is what I personally find most valuable in making a good diagram, and it often makes you fill in intermediate points that turn out useful later on.

On a related note, making a good diagram is also not a fixed process. I find it helpful to interweave working out rough sketches with filling in a precise diagram, as how to *start constructing* a problem is not always apparent at first glance.

As a related quiz: given a triangle ABC (and its circumcircle), what's the fastest way to construct the A -mixtilinear intouch point?

2 “Artificial Distinctions” and Symmetry

In geometry we often distinguish objects that feel “the same”, for configuration-related reasons. The classic example is the incircle and the three excircles, which can all be defined as a solution to the problem of finding a circle tangent to three lines. We can also specialize to just the incircle and a given excircle of a triangle, by defining both as the intersection of the corresponding angle bisector and Fact 5 circle. Notably, a kind of “duality” arises whenever we define a point by intersecting a line and a circle, as we have no way of distinguishing this intersection from the other. Hence, whenever you have a problem which “breaks” symmetry by focusing on one point amongst a (finite) tuple which should all act the same way (e.g. by focusing on only one point among a finite collection which can be defined as an intersection of circles and

lines), it may be helpful to include the other members of the tuple, to either present a more complete diagram or to understand any symmetries that appear thereafter.

Here is a recent example.

Example 2.1. Point D is selected inside acute $\triangle ABC$ so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC .

Show that line AB is tangent to the circumcircle of triangle BEM .

Here is a walkthrough, based off of Evan Chen's remarks in the AoPS thread (see post 47):

1. Construct a diagram. Along the way, dilate E by two (so that instead of the point M , we work with C).
2. The set of points which satisfy $\angle BDC = 90^\circ + \angle BAC$ form an arc on a circle. So there are really two possible choices for D . Add in the other choice, and call it D' . Similarly define F' .
3. From the diagram, what relationship seems to be true of the triangles CFE' and CDD' ? Hence what about CDF and $CD'F'$?
4. Let's now consider the point $X = DD' \cap FF'$. Can you identify this point X ? Prove whatever you find.
5. With X now giving us a foothold on the diagram, let's go back and prove that $XDFC$ is cyclic.
6. Finish the problem. (So we didn't end up needing D' and F' logically, but they were very helpful for motivation).

Please also note that it is not always the case that inequalities are necessarily symmetry breaking; there are occasionally length and angle conditions that do not satisfy certain symmetry relations which "feel intuitive". See here for an example of an inequality which is necessary for a problem to hold. So this section is mostly a general rule of thumb that is often quite effective, and outside of olympiad math you shouldn't be over-reliant on symmetries that you can't prove.

3 Get Creative!

Here is a difficult example that I would like to showcase because its methods are not completely covered within the previous sections, but the line of thinking it requires is of a similar spirit.

Example 3.1. Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD , BE , and CF are concurrent.

Here is a walkthrough. But before you begin, **I want you to spend at least 20 minutes trying to decode the lengths yourselves, and don't start this walkthrough until you feel you are completely stuck.**

1. Here is a starting point. Prove that the angles between (AB, DE) , (BC, EF) , and (CD, FA) are the same.
2. We can interpret this as follows. Put the problem on the complex plane, and let $B - A = p$, $D - C = q$, $F - E = r$ for some $p, q, r \in \mathbb{C}$. Prove that we can express $E - D = pz$, $A - F = qz$, $C - B = rz$ for some $z \in \mathbb{C}$ of norm 1.
3. Now, what does the condition

$$\sum_{\text{cyc}} \vec{B} - \vec{A} = 0$$

give you?

4. Eliminate the case $z = -1$. Interpret the case $p + q + r = 0$ using some parallelogram work. (Note we really want to be working with parallelograms here, so that we preserve the vectors themselves, not just their lengths).
5. Finish from here.

4 Problems

This problem section is split into “core problems”, which are problems which (in my opinion) reflect the themes of this lecture the closest, and “mixed bag”, which are about problems which have awkward conditions but are not necessarily fully about them (although they still may be!) Both are good practice, so work on them simultaneously. My favourite problems are labelled with ★.

4.1 Core Problems

1. Point D lies inside triangle ABC such that $\angle DAC = \angle DCA = 30^\circ$ and $\angle DBA = 60^\circ$. Point E is the midpoint of segment BC . Point F lies on segment AC with $AF = 2FC$. Prove that $DE \perp EF$.
2. Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB .

3. In triangle ABC , points E and F are taken on sides AC and AB respectively, such that $EF \parallel BC$. The angle bisector of $\angle BAC$ intersects BC at D . Point P lies on the extension of line DA beyond A , satisfying $\angle PBE = \angle ABC$ and $\angle PCF = \angle ACB$. Prove that $AP = AD$.
4. Let $ABCDE$ be a convex pentagon such that $\triangle ABE$, $\triangle BEC$, and $\triangle EDB$ are similar (with vertices in order). Lines BE and CD intersect at point T . Prove that line AT is tangent to the circumcircle of $\triangle ACD$.
5. ★ Six points are chosen on the sides of an equilateral triangle ABC : A_1, A_2 on BC , B_1, B_2 on CA and C_1, C_2 on AB , such that they are the vertices of a convex hexagon $A_1A_2B_1B_2C_1C_2$ with equal side lengths.

Prove that the lines A_1B_2 , B_1C_2 and C_1A_2 are concurrent.

6. Let ABC be an acute angled triangle and let P, Q be points on AB, AC respectively, such that PQ is parallel to BC . Points X, Y are given on line segments BQ, CP respectively, such that $\angle AXP = \angle XCB$ and $\angle AYQ = \angle YBC$. Prove that $AX = AY$.
7. Let $ABCD$ be a parallelogram. Point E lies on segment CD such that

$$2\angle AEB = \angle ADB + \angle ACB,$$

and point F lies on segment BC such that

$$2\angle DFA = \angle DCA + \angle DBA.$$

Let K be the circumcenter of triangle ABD . Prove that $KE = KF$.

8. A convex hexagon $ABCDEF$ is given where $\angle FAB + \angle BCD + \angle DEF = 360^\circ$ and $\angle AEB = \angle ADB$. Suppose the lines AB and DE are not parallel. Prove that the circumcenters of the triangles $\triangle AFE$, $\triangle BCD$ and the intersection of the lines AB and DE are collinear.
9. ★ Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK .

Show that $MK = ML$.

10. ★ Inside the inscribed quadrilateral $ABCD$ are marked points P and Q , such that $\angle PDC + \angle PCB$, $\angle PAB + \angle PBC$, $\angle QCD + \angle QDA$ and $\angle QBA + \angle QAD$ are all equal to 90° . Prove that the line PQ has equal angles with lines AD and BC .
11. Let ABP, BCQ, CAR be three non-overlapping triangles erected outside of acute triangle ABC . Let M be the midpoint of segment AP . Given that $\angle PAB = \angle CQB = 45^\circ$, $\angle ABP = \angle QBC = 75^\circ$, $\angle RAC = 105^\circ$, and $RQ^2 = 6CM^2$, compute AC^2/AR^2 .
12. Let X be a point lying in the interior of the acute triangle ABC such that

$$\sphericalangle BAX = 2\sphericalangle XBA \quad \text{and} \quad \sphericalangle XAC = 2\sphericalangle ACX.$$

Denote by M the midpoint of the arc BC of the circumcircle (ABC) containing A . Prove that $XM = XA$.

13. Let $ABCDEF$ be a convex hexagon such that $\angle A = \angle C = \angle E$ and $\angle B = \angle D = \angle F$ and the (interior) angle bisectors of $\angle A$, $\angle C$, and $\angle E$ are concurrent. Prove that the (interior) angle bisectors of $\angle B$, $\angle D$, and $\angle F$ must also be concurrent.
14. Let P be a point in the interior of quadrilateral $ABCD$ such that:

$$\angle BPC = 2\angle BAC, \quad \angle PCA = \angle PAD, \quad \angle PDA = \angle PAC$$

Prove that:

$$\angle PBD = |\angle BCA - \angle PCA|$$

15. Let $ABCDE$ be a convex pentagon such that $BC \parallel AE$, $AB = BC + AE$, and $\angle ABC = \angle CDE$. Let M be the midpoint of CE , and let O be the circumcenter of triangle BCD . Given that $\angle DMO = 90^\circ$, prove that $2\angle BDA = \angle CDE$.
16. ★ Let $ABCDE$ be a convex pentagon with $CD = DE$ and $\angle EDC \neq 2 \cdot \angle ADB$. Suppose that a point P is located in the interior of the pentagon such that $AP = AE$ and $BP = BC$. Prove that P lies on the diagonal CE if and only if $\text{area}(BCD) + \text{area}(ADE) = \text{area}(ABD) + \text{area}(ABP)$.
17. Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Suppose that the circumcentre of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC . Prove that triangle ABC is right-angled.
18. Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and

$$\angle CHS - \angle CSB = 90^\circ, \quad \angle THC - \angle DTC = 90^\circ.$$

Prove that line BD is tangent to the circumcircle of triangle TSH .

19. Let $ABCDEF$ be a convex hexagon satisfying $\overline{AB} \parallel \overline{DE}$, $\overline{BC} \parallel \overline{EF}$, $\overline{CD} \parallel \overline{FA}$, and

$$AB \cdot DE = BC \cdot EF = CD \cdot FA.$$

Let X , Y , and Z be the midpoints of $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. Prove that the circumcenter of $\triangle ACE$, the circumcenter of $\triangle BDF$, and the orthocenter of $\triangle XYZ$ are collinear.

20. ★ Given a triangle $\triangle ABC$. Denote its incenter and orthocenter by I, H , respectively. If there is a point K with

$$AH + AK = BH + BK = CH + CK$$

Show that H, I, K are collinear.

4.2 Mixed Bag

1. Let ABC be a triangle with centroid G . Points R and S are chosen on rays GB and GC , respectively, such that

$$\angle ABS = \angle ACR = 180^\circ - \angle BGC.$$

Prove that $\angle RAS + \angle BAC = \angle BGC$.

2. Let ABC be a triangle with $AC > AB$, and denote its circumcircle by Ω and incentre by I . Let its incircle meet sides BC, CA, AB at D, E, F respectively. Let X and Y be two points on minor arcs $\widehat{DF}$ and $\widehat{DE}$ of the incircle, respectively, such that $\angle BXD = \angle DYC$. Let line XY meet line BC at K . Let T be the point on Ω such that KT is tangent to Ω and T is on the same side of line BC as A . Prove that lines TD and AI meet on Ω .
3. Given a triangle ABC satisfying $AC + BC = 3 \cdot AB$. The incircle of triangle ABC has center I and touches the sides BC and CA at the points D and E , respectively. Let K and L be the reflections of the points D and E with respect to I . Prove that the points A, B, K, L lie on one circle.
4. Let O be the circumcenter and H the orthocenter of an acute triangle ABC . Show that there exist points D, E , and F on sides BC, CA , and AB respectively such that

$$OD + DH = OE + EH = OF + FH$$

and the lines AD, BE , and CF are concurrent.

5. Let $ABCD$ be a cyclic quadrilateral with $\angle BAD < \angle ADC$. Let M be the midpoint of the arc CD not containing A . Suppose there is a point P inside $ABCD$ such that $\angle ADB = \angle CPD$ and $\angle ADP = \angle PCB$.

Prove that lines AD, PM , and BC are concurrent.

6. Let ABC be a triangle with $AB = AC$, and let M be the midpoint of BC . Let P be a point such that $PB < PC$ and PA is parallel to BC . Let X and Y be points on the lines PB and PC , respectively, so that B lies on the segment PX , C lies on the segment PY , and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic.
7. Consider triangle ABC with orthocenter H . Let points M and N be the midpoints of segments BC and AH . Point D lies on line MH so that $AD \parallel BC$ and point K lies on line AH so that $DNMK$ is cyclic. Points E and F lie on lines AC and AB such that $\angle EHM = \angle C$ and $\angle FHM = \angle B$. Prove that points D, E, F and K lie on a circle.
8. Let ABC be an acute triangle. Let M be the midpoint of side BC , and let E and F be the feet of the altitudes from B and C , respectively. Suppose that the common external tangents to the circumcircles of triangles BME and CMF intersect at a point K , and that K lies on the circumcircle of ABC . Prove that line AK is perpendicular to line BC .

9. ★ In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be the point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$.

Prove that points P, Q, P_1 , and Q_1 are concyclic.

10. On sides AB and AC of an acute-angled, non-isosceles triangle ABC , points P and Q are chosen such that the center O_9 of the nine-point circle of $\triangle ABC$ is the midpoint of segment PQ . Let O be the circumcenter of $\triangle ABC$. On the ray OP beyond P , segment PX is marked such that $PX = AQ$. On the ray OQ beyond Q , segment QY is marked such that $QY = AP$. Prove that the midpoint of side BC , the midpoint of segment XY , and the point O_9 are collinear.

11. Let ABC be a scalene triangle, and let D be a point on side BC satisfying $\angle BAD = \angle DAC$. Suppose that X and Y are points inside ABC such that triangles ABX and ACY are similar and quadrilaterals $ACDX$ and $ABDY$ are cyclic. Let lines BX and CY meet at S and lines BY and CX meet at T . Prove that lines DS and AT are parallel.
12. ★ Let ABC be a triangle and A', B', C' be the midpoints of BC, CA, AB respectively. Let P and P' be points in plane such that $PA = P'A', PB = P'B', PC = P'C'$. Prove that all PP' pass through a fixed point.
13. In acute triangle ABC , segments AD, BE , and CF are its altitudes, and H is its orthocenter. Circle ω , centered at O , passes through A and H and intersects sides AB and AC again at Q and P (other than A), respectively. The circumcircle of triangle OPQ is tangent to segment BC at R . Prove that $\frac{CR}{BR} = \frac{ED}{FD}$.
14. ★ Perimeter of triangle ABC is 4. Point X is marked at ray AB and point Y is marked at ray AC such that $AX = AY = 1$. Line segments BC and XY intersect at point M . Prove that perimeter of one of triangles ABM or ACM is 2.
15. Let $ABCD$ be a cyclic quadrilateral whose sides have pairwise different lengths. Let O be the circumcenter of $ABCD$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at B_1 and D_1 , respectively. Let O_B be the center of the circle which passes through B and is tangent to $\overline{AC}$ at D_1 . Similarly, let O_D be the center of the circle which passes through D and is tangent to $\overline{AC}$ at B_1 . Assume that $\overline{BD_1} \parallel \overline{DB_1}$. Prove that O lies on the line $\overline{O_B O_D}$.
16. In $\triangle ABC$, points D, E , and F lie on sides BC, CA , and AB , respectively, such that each of the quadrilaterals $AFDE, BDEF$, and $CEFD$ has an incircle. Prove that the inradius of $\triangle ABC$ is twice the inradius of $\triangle DEF$.
17. Let $ABCD$ be a convex quadrilateral. The perpendicular bisectors of its sides AB and CD meet at Y . Denote by X a point inside the quadrilateral $ABCD$ such that $\angle ADX = \angle BCX < 90^\circ$ and $\angle DAX = \angle CBX < 90^\circ$. Show that $\angle AYB = 2 \cdot \angle ADX$.
18. Point P lies in the interior of a triangle ABC . Lines AP, BP , and CP meet the opposite sides of triangle ABC at A', B' , and C' respectively. Let P_A the midpoint of the segment joining the incenters of triangles BPC' and CPB' , and define points P_B and P_C analogously. Show that if

$$AB' + BC' + CA' = AC' + BA' + CB'$$

then points P, P_A, P_B , and P_C are concyclic.

19. ★ Denote H and I as the orthocenter and incenter, respectively, of triangle $\triangle ABC$. Let M be the midpoint of $\overline{BC}$. Prove that $\angle HIM = 90^\circ$ if and only if $AB + AC = 2BC$.
20. Let ABC be a triangle with incenter I , and let D be a point on line BC satisfying $\angle AID = 90^\circ$. Let the excircle of triangle ABC opposite the vertex A be tangent to $\overline{BC}$ at A_1 . Define points B_1 on $\overline{CA}$ and C_1 on $\overline{AB}$ analogously, using the excircles opposite B and C , respectively. Prove that if quadrilateral $AB_1A_1C_1$ is cyclic, then $\overline{AD}$ is tangent to the circumcircle of $\triangle DB_1C_1$.
21. ★ In triangle ABC let O and H be the circumcenter and the orthocenter. The point P is the reflection of A with respect to OH . Assume that P is not on the same side of BC as A . Points E, F lie on AB, AC respectively such that $BE = PC, CF = PB$. Let K be the intersection point of AP, OH . Prove that $\angle EKF = 90^\circ$.
22. In triangle ABC , M, N and P are midpoints of sides BC, CA and AB . Point K lies on segment NP so that AK bisects $\angle BKC$. Lines MN, BK intersect at E and lines MP, CK intersect at F . Suppose that H be the foot of perpendicular line from A to BC and L the second intersection of circumcircle of triangles AKH, HEF . Prove that MK, EF and HL are concurrent.

23. A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.

24. Let ABC be an equilateral triangle. Let A_1, B_1, C_1 be interior points of ABC such that $BA_1 = A_1C$, $CB_1 = B_1A$, $AC_1 = C_1B$, and

$$\angle BA_1C + \angle CB_1A + \angle AC_1B = 480^\circ$$

Let BC_1 and CB_1 meet at A_2 , let CA_1 and AC_1 meet at B_2 , and let AB_1 and BA_1 meet at C_2 . Prove that if triangle $A_1B_1C_1$ is scalene, then the three circumcircles of triangles AA_1A_2 , BB_1B_2 and CC_1C_2 all pass through two common points.

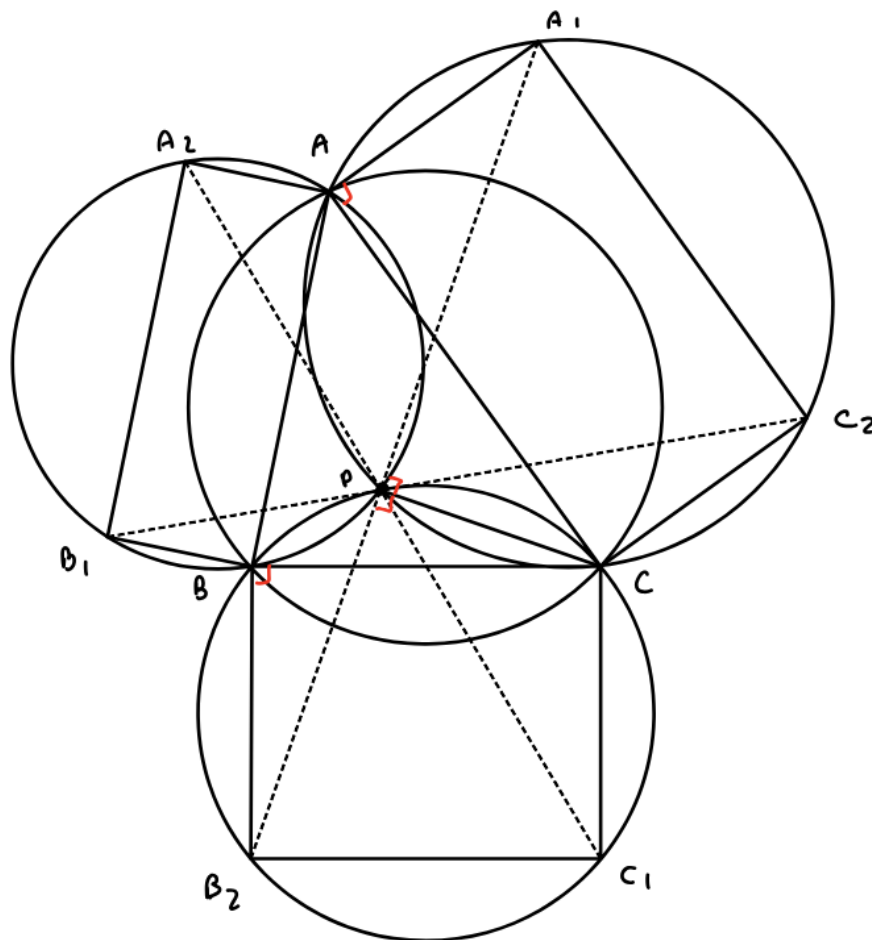
5 Solutions to Walkthroughs

Example 5.1. Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

Proof. The angle condition is equivalent to (ABB_1) , (BCC_1) , (CAA_1) intersecting at some point. Call it P . Then $\angle CPB_2 + \angle CPA_1 = \angle CBB_2 + \angle CAA_1 = 180^\circ$. So P is the desired intersection. $\square$

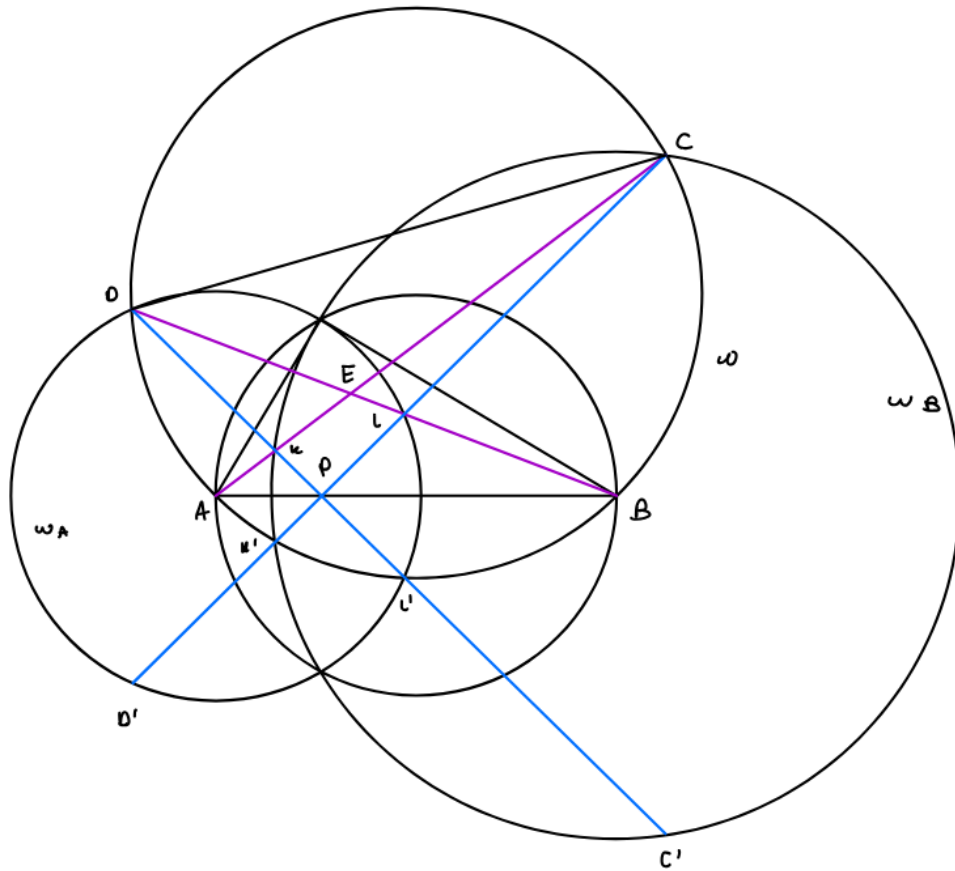


Example 5.2. Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

Proof. Redefine P as the point satisfying that $AP \cdot AB = AD^2$. Note that $BP \cdot BA = AB^2 - AD^2 = BC^2$, so P is both the inverse of B with respect to ω_A and A with respect to ω_B . Notably we have $\angle APD = \angle ADB = \angle ACB = \angle CPB$, so P is the (unique; construct P by reflecting C across AB to make C' and intersecting $C'D$ with AB) point satisfying the given conditions.

Now, note that $AD/AB = (AD/AB)^2$, so $AD/BD = (AD/BC)^2 = (AE/BE)^2$. Hence DE is the E -symmedian of EAB , and hence DE is the E -median of ECD , as desired.

In fact, for experienced solvers, this last condition can be back-constructed to directly identify P . This is what MarkBcc168 does in his solution in the AoPS thread (see post 29). $\square$

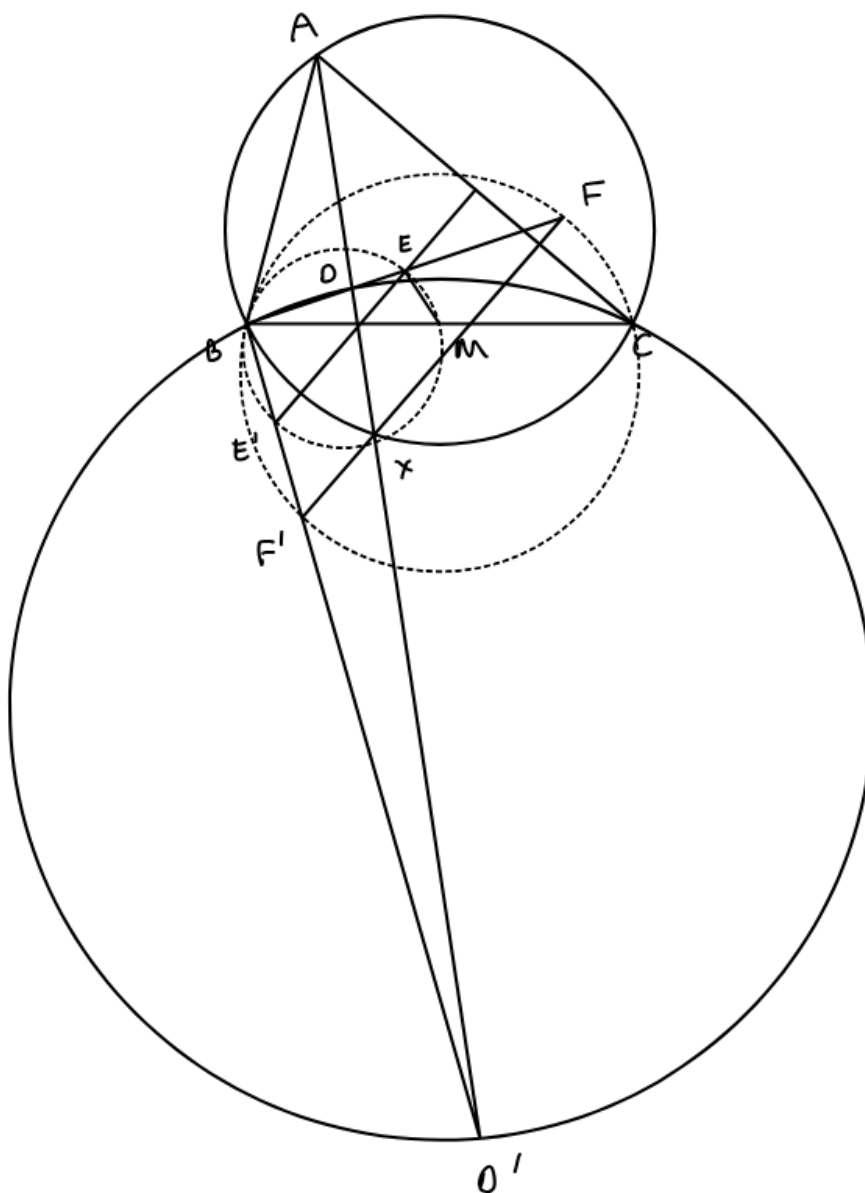


Example 5.3. Point D is selected inside acute $\triangle ABC$ so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC .

Show that line AB is tangent to the circumcircle of triangle BEM .

Proof. Dilate E by a factor of 2 to get F ; evidently it suffices to prove that (BFC) is tangent to AB . Define X to be the second intersection of AD and (ABC) . Note that $BX \parallel AC$, so BX and AC share perpendicular bisectors. Hence $BF \perp AC$.

Now, note that $\angle FDC = 180^\circ - \angle BDC = 90^\circ - \angle A = 90^\circ - \angle ACX = \angle FCX$, so $DFCX$ is cyclic. Now $\angle BFC = 180^\circ - \angle AXC = 180^\circ - \angle ABC$, so (BFC) is tangent to AB , as desired. $\square$



Example 5.4. Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD , BE , and CF are concurrent.

Proof. Denote $x = \angle A - \angle D$. The angle made by AB and DE is $360^\circ - \angle A - \angle E - \angle F = 360^\circ + x - \angle A - \angle C - \angle E$. This is symmetric, hence it is also equal to the angle made by BC and EF , and the angle made by CD and FA . One can readily check in fact the directed angles are equal. Hence if we consider the configuration on the complex plane and let $\vec{B} - \vec{A} = p$, $\vec{D} - \vec{C} = q$, $\vec{F} - \vec{E} = r$, then $\vec{D} - \vec{E} = pz$, $\vec{F} - \vec{A} = qz$, and $\vec{C} - \vec{B} = rz$ for some $z \in \mathbb{C}$ of norm one.

Now, since $\sum_{\text{cyc}} \vec{B} - \vec{A} = 0$, we have that $(p + q + r)(z + 1) = 0$. The case $z = -1$ corresponds to when the opposite sides of the hexagon are parallel, in which case the intersection point is the shared midpoint of AD , BE , and CF (for a proof, use vectors). We now restrict to the case where $z \neq -1$, hence $p + q + r = 0$.

In this case, define X such that $ABXF$ is a parallelogram. Since $\vec{X} - \vec{F} = p$ and $\vec{F} - \vec{E} = r$, we have $\vec{E} - \vec{X} = q$, and hence $EXCD$ is also a parallelogram! Similarly construct Y and Z such that $YABC$, $YDEF$, $ZBCD$, and $ZEFA$ are parallelograms. We now have that $CX = XF = FY = YC = AB$, so $CXFY$ is a rhombus. Similarly $BZEX$ and $AYDZ$ are rhombi, so AD, BE, CF concur at the circumcenter of XYZ . $\square$

